HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** ETL pipelines for extracting and transforming agent telemetry data, processing approximately 1.5 million records daily from various sources.
- Developed and maintained **Java Spring Boot** microservices to support the agent metrics platform, focusing on REST API development for data ingestion and retrieval.
- Engineered **SQL** queries and database schemas on **PostgreSQL** for the semantic layer, ensuring efficient data retrieval and supporting complex analytical needs for over 50 distinct metrics.
- Contributed to the technical design and execution of core components for the Agent Measurement & Forecasting (AMF) platform, focusing on data pipeline development.
- Implemented schema contract enforcement using **CI/CD** gates and automated data quality checks, ensuring data integrity for all published metrics within the platform.
- Collaborated with cross-functional teams to define and serve governed metrics through the **MetricsFlow** semantic layer, supporting downstream consumers and the Product Data Store.
- Containerized microservices using **Docker** and managed deployments on **AWS EC2** instances, ensuring reliable operation within the AWS cloud environment.

SKILLS

Languages – Python, SQL, Java

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, dbt

Data Technologies – Apache Iceberg, PostgreSQL, MySQL, Oracle, MetricsFlow

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Processing – ETL, Data Warehousing, Data Modelling, API Development (REST, gRPC)

Methodologies – Agile, Scrum, CI/CD

PROJECTS

Agent Telemetry Data Pipeline (HCL Technologies)

- Built a scalable **Python** data pipeline to ingest raw agent telemetry, processing over 1 million events daily and landing data into **Apache Iceberg** tables.
- Developed **SQL** transformations using **dbt** to curate and aggregate raw data into auditable metrics, ensuring consistency and rigor across the platform.
- Implemented data validation checks and logging within the pipeline to monitor data quality and identify discrepancies before metrics were published.

Metrics Semantic Layer Service (HCL Technologies)

- Designed and developed a **Java Spring Boot** service exposing REST APIs to serve governed metrics from the semantic layer to various internal applications.
- Integrated with **PostgreSQL** to efficiently query and retrieve metric definitions and associated data, supporting low-latency access for downstream consumers.
- Wrote comprehensive unit tests using **JUnit** to ensure the reliability and correctness of the API endpoints and data retrieval logic.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)